

BY 531: Quantitative Biology and Data Analytics

Course Code	BY 531
Course Name	Quantitative Biology and Data Analytics
L-T-P-C	3-0-0-3
Prerequisites	IC136 - Understanding Biotechnology & its Applications or Consent of faculty member
Intended for	Core for M.Tech Biotechnology and elective for B.Tech, M.Sc/M.Tech and PhD candidates
Approval	42nd Senate (Jan 2024), in the M.Tech Biotechnology curriculum; 54th BoA (May 2024)

Course Contents

- **Module 1:** Introduction to statistics and probability theory (distributions, mean, standard deviations, variance, etc.), basic principles of statistical inference, statistical models, linear models, basic bayesian methods, confidence interval estimation and hypothesis testing, including P-values, significance level, power, sample size, and two types of errors. (4 hours)
- **Module 2:** Linear algebra and matrices, inference for high dimensional data, distance and dimension reduction, PCA, t-SNE, etc., machine learning, and batch effects. (8 hours)
- **Module 3:** Data visualization, exploratory data analysis, and robust summaries. (4 hours)
- **Module 4:** Mathematical models for biological processes, basic calculus, logistic regression, bifurcation and steady state analysis in population growth models, epidemic models, and predator-prey models. (6 hours)
- **Module 5:** Image processing in MATLAB: types of image - binary, grayscale, & color; image thresholding and segmentation; particle tracking, and z-stack projections. (8 hours)

Text Books

1. Brian Munsky et. al., **Quantitative Biology: Theory, Computational Methods, and Models**.
2. Whitlock, Michael C.; Schluter, Dolph., **The Analysis of Biological Data**, 2nd Edition, Freeman, W. H. & Company, 2014

References

1. Data Science for Biology, Emily Ren et. al., Online materials, etc.